PHANIDHAR AKULA

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Simulation Systems, Digital Twins & Geospatial Computing • U.S. work authorization: OPT pending for Oct 2026; STEM OPT eligible

EDUCATION

Miami University | M.S. in Computer Science — *Aug 2024 – Aug 2026*
Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Graduate Assistantship
Thesis – SimForge: A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (defended July 2026; OhioLINK ETD)
Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science — *Aug 2020 – May 2024*
Hyderabad, India • CGPA: 8.16/10.00 • Two peer-reviewed publications in CNN-based soil classification and LDA topic modeling

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE — *Aug 2025 – May 2026*
Research group of Dr. DJ Rao
- Graduate research in the simulation/digital-twin group; lead developer on Cityscape, integrated into the SimForge thesis pipeline.
- Co-administered the Grand Challenges Scholars Program (GCSP), mentoring undergraduate research while supporting faculty-led research initiatives and program operations.

Smart Pathfinders Overseas Education | Software Engineer Intern — *May 2023 – Jul 2024*
- Built an internal staff management and organizational hierarchy platform for an early-stage startup.
- Shipped production React/TypeScript features, collaborating with product and design on iterative releases while resolving UI regressions and edge cases identified through post-release feedback.

Independent Clients | Freelance Software Engineer — *Aug 2023 – May 2024*
- Delivered end-to-end full-stack features for three early-stage startup clients under NDA.
- Built React/TypeScript frontends and Python/MySQL backend services exposed through REST APIs, maintaining AWS infrastructure while balancing performance, security, and maintainability.

SELECTED PROJECTS

SimForge | Master's Thesis — *Aug 2024 – Jul 2026*
Python, OpenMP, multiprocessing, SUMO, MATSim, DTALite, Docker, Apptainer, SLURM, OSC HPC (Pitzer/Cardinal), OSM, US Census PUMS
- Built an open-source reproducible benchmarking framework integrating SUMO, MATSim, and DTALite under a canonical dataset pipeline for fair cross-simulator evaluation; released under Apache-2.0 with a citable Zenodo DOI (10.5281/zenodo.21385983).
- Reduced shared-route pre-routing from 141.9 hours to 7.1 hours (20× cold-cache, 228× warm) through deterministic route caching and 16-way parallelism; deployed on OSC Pitzer and Cardinal HPC clusters.
- Benchmarked three cities across demand tiers up to 500K trips, revealing regime-dependent cross-engine behavior and up to 96% travel-time divergence under saturation.

Cityscape | Open-Source Population Synthesis and Geospatial Demand Generation — *Aug 2025 – Jun 2026*
C++, OpenMP, OSM/Geofabrik, US Census PUMS, OSC HPC (Pitzer)
- Authored three merged C++ pull requests (github.com/raodj/cityscape, PRs #1–#3) to an open-source PUMS-driven population synthesizer and trip-demand generator. Fixed OSM building-classification and orphan-ring logic across five metros, collapsing synthetic-home ratios (e.g. LA ~70% → ~5%) and removing ~1.25M stranded-population artifacts, impacting over one million building records.
- Exposed search-radius CLI flags and ran a 4-city, 16-combination HPC parameter sweep on OSC Pitzer, raising worst-case calibration R^2 from 0.812 to 0.917. Contributions integrated into the SimForge thesis demand pipeline.

LumiAI | Production AI Tutoring Platform – studywithlumi.com — *Nov 2025 – Present*
React, TypeScript, LLMs, Supabase (PostgreSQL, Auth, Storage), Vercel Serverless
- Solo-architected, built, deployed, and now maintain a production AI tutoring platform serving 60+ active students.
- Implemented document-grounded AI chat, quiz and flashcard generation, SM-2 spaced repetition, voice mode, Google OAuth, row-level security, server-side API key isolation, and an admin analytics dashboard.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Java
HPC & Systems: Linux, Git, OpenMP, multiprocessing, SLURM, OSC Pitzer/Cardinal, Docker, Apptainer, GitHub Actions, GPU-accelerated simulation
Engineering & Evaluation: Benchmarking frameworks, reproducibility auditing, statistical validation (95% CIs), pytest, mutation testing, code review
AI/ML & Data: PyTorch, TensorFlow, LLMs, CNNs, LDA/topic modeling, FastAPI, REST APIs, GraphQL, MySQL, PostgreSQL, AWS
Simulation & Geospatial: SUMO, MATSim, DTALite, digital twins, agent-based and traffic simulation, OSM/Geofabrik, US Census PUMS pipelines, network modeling

PUBLICATIONS

- **Akula, P.**, & Rao, D. M. Attributable Cross-Simulator Benchmarking for Urban Traffic Simulation: Reproducibility, Consistency, and Regime-Dependent Divergence. IEEE Transactions on Intelligent Transportation Systems (under review, first author, 2026).
- **Akula, P.**, Stojanovski, S., & Rao, D. M. CITYSCAPE: A Digital Twin Framework for Temporospatial Modeling of Work-Commute Dynamics and Roadway Infrastructure Demand. Frontiers (under review, first author, 2026).

LEADERSHIP

- President, Graduate Students of Color Association (104 members; Certificate of Appreciation, 2026), and elected Board Member, Graduate Student Council, Miami University (Aug 2025 – May 2026). Led recruitment, events, and graduate student advocacy.